



VocalIQ

Detailed Client Pitch, Tech-Stack Cost & Profit Model

Agentic Voice AI SaaS — har feature ka detail, har third-party service ki pricing + use-case, 6 tech-stack combinations, owner ka monthly infra kharcha (100 → 5000 clients), aur revenue / net-profit projections. **Confidential — for internal + client presentation.**

📌 Executive Summary — ek nazar mein

~\$0.035–0.11

COST / AI CALL MINUTE

85–90%

GROSS MARGIN (BYOK MODEL)

\$0.15–0.25

CHARGE / MIN (MANAGED)

30+

THIRD-PARTY INTEGRATIONS

Business model (kaise paisa banta hai): VocalIQ do tareeke se kamata hai — (1) **Subscription** (har client monthly plan pay kare), aur (2) **Managed minutes markup** (client humari AI keys use kare to hum cost pe markup lagayein). Sabse smart: **BYOK** (client apni AI keys laaye) → aap sirf platform fee lete ho, AI ka kharcha client bharta hai → **90%+ margin**. White-label reseller layer se aap bina sales kiye scale karte ho (resellers apne clients laate hain, aap unse fee + revenue-share lete ho).

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VocallQ Features — detail explanation + use case

Har feature: kya karta hai (Hinglish) + kis business situation mein kaam aata hai.

FEATURE	KYA KARTA HAI (DETAIL)	USE CASE
 Provider Router	Saare LLM/STT/TTS/telephony calls ek jagah se route — naya vendor add karna sirf config, code nahi.	Aaj OpenAI mehenga ho gaya? 1 setting se Gemini pe switch — clients ko pata bhi nahi chalega.
Multi-LLM brain	GPT, Claude, Gemini, Grok ya 100+ models (OpenRouter) me se agent ka dimaag chuno.	Simple FAQ ke liye sasta Gemini Flash; complex sales ke liye Claude Opus.
BYOK / Managed keys	Client apni provider keys laaye (thin platform fee) ya humari credits use kare (marked-up minutes).	Bade clients BYOK (cost bachaye), chhote clients managed (convenience).
 Real-time voice loop	Agent live sunta → sochta → bolta (STT→LLM→TTS), phone-call jaisa natural, low latency.	Inbound support, outbound sales — insaan jaisa AI caller 24/7.
Inbound + Outbound calls	PSTN pe calls uthaye bhi aur kare bhi; concurrency + rate caps built-in.	Din bhar inbound queries handle + raat ko outbound campaign.
Phone number buy	App ke andar hi number search/buy/release (Twilio/Telnyx/Plivo).	Client ko naya local/toll-free number chahiye → 2 click mein mil jaye.
SIP trunk	Enterprise apna SIP/PBX (Asterisk/Kamailio) jod sake.	Bade call-center clients apni existing telephony reuse karein.
Web call widget	Website pe "Call us" button → browser se seedha AI voice call (no phone).	E-commerce site pe visitor turant AI se baat kare — lead capture.
WhatsApp Calling	WhatsApp par AI agent voice call — inbound FREE, consented outbound with permission governor.	India/emerging markets jahan WhatsApp #1 hai — sabse sasta channel.
Messenger Calling	Facebook Page par AI voice call — Meta side \$0, sirf AI cost.	Social-commerce brands jinke customers Messenger pe hain.
Spam-shield (STIR/SHAKEN)	Outbound calls "Spam Likely" na dikhein — caller reputation/branded ID.	Outbound pickup-rate badhe → zyada connected calls → zyada sales.
 Drag-drop flow builder	Visual node canvas — coding ke bina poori conversation design.	Non-technical client khud apna call-flow bana le.
Templates + Squads	Ready agent templates + kai specialized agents ki team (aapas mein handoff).	"Reception agent" query samjhe → "Sales agent" ko transfer kare.
Knowledge base (RAG)	Agent aapke docs/website se jawab de (embeddings + vector search).	Product catalog upload → agent accurate product answers de.
Memory	Agent ko contact ki past baatein yaad rahein (opt-in, GDPR-erase).	Repeat customer ko "welcome back" + last order context.
Voice cloning + 50+ languages	Brand ki apni awaaz + real-time multilingual translation call ke beech.	Ek agent Hindi, English, Tamil — sab bole; brand voice consistent.
 Bulk campaigns	Hazaaron outbound calls schedule, pacing/concurrency caps, auto-retry failed.	10,000 leads ko ek din mein AI se call — human se 100x fast.
Form-to-Call 	Form submit hote hi submitter ko seconds mein AI call.	Website lead form bhara → 30 sec mein call → 8x higher conversion.
Appointments + Callbacks	Google Calendar 2-way sync + scheduled callback auto-dial.	Clinic/salon booking; "busy hoon, 5 baje call karo" → auto.
A/B experiments	Do scripts/agents ki tulna — data se winner chuno.	Kaunsi opening line zyada appointments deti hai — measure.
 Post-call intel	Har call ka AI summary, key points, action items.	Manager 500 calls na sune — bas summaries padhe.
Sentiment + QA scoring	Caller ka mood + har call ka automatic quality score (rubric).	Unhappy customers auto-flag; agent quality objectively measure.
Transcription + semantic search	Recording → transcript; sab calls mein matlab-based search.	"Kis-kis ne refund manga tha" — 1 search mein saari calls.
Revenue attribution	Kaunsi call se deal/paisa aaya — outcomes analytics.	Client ko ROI dikhe → renewal pakka.
 CRM + integrations	HubSpot live, Salesforce/Zendesk ready, Slack, n8n (400+ apps).	Call ke baad lead auto CRM mein + Slack notification.
Workflow automations	Event → action rules + run-logs + retry.	"Call missed → SMS bhejo → 1 ghante baad callback".
Video avatar + voice biometrics	Talking-head video agent (HeyGen/D-ID/Tavus) + awaaz se identity verify.	Premium add-on: video kiosk agent; bank-grade voice auth.
 Per-call cost + billing	Har call ka STT+LLM+TTS+phone cost live; Stripe subscriptions + wallet + promo credits.	Aapko exact margin dikhe; client ko transparent usage.
 White-label + reseller	Custom domain + branding; reseller sub-tenants + apna markup pricing.	Agency aapke platform ko apni brand se apne 50 clients ko bech'e.

 Multi-tenant + compliance	Har client ka data 100% alag (RLS); fraud/anti-abuse; data residency; SSO/SAML; audit logs.	Enterprise + GDPR/regulated clients bharose se onboard.
 Agent Desk + API/SDK	Human live call takeover; public metered API + SDK + in-app "Try it".	Complex call pe human takeover; developers apne app mein embed karein.

Poori feature list (100+ modules, category-wise) project ke [docs/PROJECT-FEATURES-EXPLAINED](#) mein hai — ye top business-facing features hain.

2 Third-party services — pricing, free tier & use case

VocallQ provider-agnostic hai — ye vendors router-seam ke peeche plug hote hain. Prices July 2026 (official pages).  = code mein active,  = planned/gated.

LLM — agent ka dimaag (per 1M tokens)

PROVIDER	STATUS	INPUT / OUTPUT	FREE TIER	USE CASE
Anthropic Claude		Opus 5 \$5/\$25 · Sonnet 5 \$3/\$15 · Haiku 4.5 \$1/\$5	trial credits	Best reasoning — complex sales/support agent.
OpenAI GPT		GPT-5.6 Terra \$2.5/\$15 · Luna \$1/\$6 · embeddings \$0.02–0.13	—	Alt brain + RAG embeddings (knowledge base).
Google Gemini		Flash \$0.30/\$2.50 · Flash-Lite \$0.10/\$0.40	 free tier	Sasta bulk — high-volume simple calls.
xAI Grok		Grok-4.5 \$2/\$6 · 4.1-Fast \$0.20/\$0.50	~\$175/mo credits	Alt/fallback brain.
OpenRouter		pass-through + 5–5.5% fee	50 req/day	Ek key se 100+ models test/switch.

 **Smart:** Prompt-caching + batch = **50–90% saste**. High-volume ke liye Gemini Flash-Lite (\$0.10/1M) → LLM cost ~\$0.005/min tak gir jaata hai.

Voice — STT (kaan) & TTS (awaaz)

PROVIDER	TYPE	PRICE	FREE	USE CASE
Deepgram	STT 	Nova-3 \$0.0048/min	\$200 credits	Caller ki awaaz → text, accurate + fast.
AssemblyAI	STT 	\$0.0025–0.0075/min	\$50 credits	Alt STT — cheaper streaming.
ElevenLabs	TTS 	~\$0.006–0.012/min (Flash v2.5)	10K chars/mo	Best natural voice + cloning.
Cartesia	TTS 	~\$0.006/min (40ms latency)	20K chars/mo	Fastest — real-time low latency.
PlayHT	TTS 	~\$0.005/min (\$30/1M chars)	trial	Cheapest high-volume TTS.
OpenAI Realtime (S2S)		~\$0.04–0.10/min	—	Direct voice→voice, lowest latency premium.
Gemini Live (S2S)		~\$0.023/min		Sasta S2S option.

Telephony · SIP · Virtual numbers

PROVIDER	PSTN OUT (US)	PSTN IN	NUMBER/MO	SIP TRUNK	USE CASE
Twilio 	\$0.014/min	\$0.0085	\$1.15 local	\$0.010	Most reliable/mature — enterprise.
Telnyx 	~\$0.007/min	~\$0.005	\$1.00 (→\$0.25)	~\$0.005	Cheap + good — default choice.
Plivo 	\$0.0115/min	\$0.0055	\$0.50 local	\$0.0046	Cheapest SIP + numbers.
WhatsApp Calling 	inbound FREE	FREE	—	—	Outbound per-country (India ~\$0.01/min); cheapest inbound.
Messenger Calling 	\$0 (Meta)	\$0	—	—	Sirf AI cost — koi telephony fee nahi.

Real-time media, storage, infra

SERVICE	KYA KE LIYE	PRICE	FREE
LiveKit Cloud 	Har call ka WebRTC audio room (core dependency)	Ship \$50/mo (150K min) · Scale \$500/mo (1.5M min) · overage \$0.0005/min	5K min/mo
Cloudflare R2 	Call recordings storage (egress FREE)	\$0.015/GB-mo	10 GB
Supabase / Postgres 	Main multi-tenant DB (RLS)	Pro \$25/mo → Team \$599; ya self-host	500MB
Redis 	Queues (BullMQ) + cache + rate-limit	self-host free / Cloud ~\$10–50/mo	
Cloudflare for SaaS 	Reseller custom domains	\$0.10/hostname/mo	100 hostnames

Platform SaaS — billing, auth, email, ops

SERVICE	KYA KE LIYE	PRICE	FREE
Stripe 	Subscriptions + payments + reseller rebilling	2.9%+\$0.30 · Billing 0.5–0.7% · Connect 0.25%	—
Resend 	Transactional email (OTP, notif)	Pro \$20/mo (50K)	3K/mo
Sentry + PostHog 	Error monitoring + product analytics	\$26 + usage · \$0.00005/event	5K err / 1M events
Doppler 	Secrets manager	\$0–21/user	1–3 users
Vercel 	Web (Next.js) hosting	Pro \$20/user	Hobby

WorkOS ➡ SOON	Enterprise SSO/SAML	\$125/connection/mo	1M MAU
HeyGen/Tavus/D-ID ➡ SOON	Video avatar (premium add-on)	~\$0.32-3/min	trial
Google Calendar/Sheets ➡ SOON	Appointment + form sync	FREE (OAuth quota)	✓

3 Ek AI call minute ka cost — breakdown

Ye "unit economics" hai — isi pe poora margin banta hai. Har layer stack hoti hai.

LAYER	BUDGET STACK	BALANCED STACK	PREMIUM STACK
STT (kaan)	Deepgram \$0.0048	Deepgram \$0.0048	Deepgram \$0.0048
LLM (dimaag)	Gemini Flash-Lite ~\$0.005	Haiku/GPT-mini ~\$0.012	Opus/GPT-5.6 ~\$0.04
TTS (awaaz)	Cartesia/PlayHT ~\$0.006	ElevenLabs Flash ~\$0.010	ElevenLabs Multi ~\$0.018
Telephony	Plivo ~\$0.007	Telnyx ~\$0.010	Twilio ~\$0.014
Media (LiveKit)	self-host ~\$0.001	Cloud ~\$0.002	Cloud ~\$0.003
TOTAL / minute	≈ \$0.024/min	≈ \$0.036/min	≈ \$0.080/min

⚠ **Note:** WhatsApp/Messenger inbound calls sabse saste — telephony \$0, sirf AI (~\$0.02/min). S2S (OpenAI Realtime) premium: ~\$0.05–0.10/min (STT+TTS+LLM ek hi model mein). Prices quarterly badalte hain — margin ke liye re-verify.

4 6 Tech-stack Combinations — kaunsa kab

COMBO	STACK	\$/MIN	QUALITY	KAB USE KAREIN
1. Ultra-Budget max margin	Deepgram + Gemini Flash-Lite + PlayHT + Plivo + self-host LiveKit	~\$0.024	Good	High-volume simple calls, price-sensitive markets.
2. Budget	Deepgram + Gemini Flash + Cartesia + Telnyx + self-host LiveKit	~\$0.030	Good+	Default sasta — SMB clients.
3. Balanced recommended	Deepgram + Haiku/GPT-mini + ElevenLabs Flash + Telnyx + LiveKit Ship	~\$0.036	Great	Best quality/price — most clients.
4. Premium	Deepgram + Sonnet/GPT-5.6 + ElevenLabs Multi + Twilio + LiveKit	~\$0.080	Excellent	Enterprise, high-value sales calls.
5. Ultra-Premium (S2S)	OpenAI Realtime + ElevenLabs + Twilio + LiveKit Scale	~\$0.10	Best (lowest latency)	Flagship demo, VIP clients.
6. WhatsApp-first cheapest	Inbound WhatsApp (telephony free) + Gemini Flash + Cartesia	~\$0.018	Good+	India/emerging — WhatsApp-heavy audience.

Aap ek hi platform pe per-tenant/per-agent alag combo chala sakte ho — router runtime pe switch karta hai. Managed clients ko Balanced do; BYOK clients apni marzi.

5 Owner (aapka) monthly kharcha — scale ke saath

Assumptions: avg 1,500 call-min/client/month, Balanced stack \$0.036/min. Do models — (A) BYOK-first (client AI cost bhare), (B) All-managed (aap sab bhare).

A Fixed platform infra (chahe kitne bhi clients, ye base kharcha)

Component	100 Clients	500 Clients	1,000 Clients	5,000 Clients
Hosting (api/voice/workers)	\$80	\$300	\$700	\$2,500
Database (Postgres)	\$25	\$120	\$400	\$1,500
LiveKit media (or self-host)	\$50	\$200	\$500	\$1,500
Redis + R2 storage	\$20	\$80	\$200	\$900
Observability + Email	\$50	\$115	\$290	\$950
Misc (Vercel, Doppler, CF, domains)	\$60	\$85	\$150	\$400
Fixed total / month	~\$285	~\$900	~\$2,240	~\$7,750

B Model A — BYOK-first (owner sirf infra + telephony)

Clients	Fixed	Telephony*	Owner Cost/Mo
100	\$285	\$1,800	~\$2,100
500	\$900	\$9,000	~\$9,900
1,000	\$2,240	\$18,000	~\$20,200
5,000	\$7,750	\$90,000	~\$97,750

*Telephony ~\$0.012/min. AI (STT/LLM/TTS) client ki keys pe → owner ka kharcha nahi.

B Model B — All-managed (owner sab bhare, phir markup)

Clients	Fixed	Minutes (\$0.036)	Owner Cost/Mo
100	\$285	\$5,400	~\$5,700
500	\$900	\$27,000	~\$27,900
1,000	\$2,240	\$54,000	~\$56,240
5,000	\$7,750	\$270,000	~\$277,750

Managed mein aap client se markup lete ho (agla section) — cost zyada, par revenue bhi zyada.

Takeaway: BYOK-first model rakho → owner ka kharcha 100 clients pe sirf ~\$2,100/mo, 1000 pe ~\$20K/mo. AI ka bhaari cost client bharta hai, aap high-margin platform fee lete ho. Managed sirf convenience upsell.

6

Revenue packages — client kya pay kare

PLAN	PRICE/MO	INCLUDED	TARGET
Starter	\$49	1 agent, 500 managed min (ya BYOK), core features	Solo/small business.
Growth popular	\$149	3 agents, 2,000 min, campaigns + analytics	Growing SMB.
Pro	\$399	10 agents, 6,000 min, all channels + integrations	Established business.
Scale	\$999	Unlimited agents, 20,000 min, priority + SSO	Enterprise.
Reseller / White-label	\$1,999+	Custom domain, sub-tenants, markup engine + revenue-share	Agencies.
+ Managed minutes overage: included ke baad \$0.15–0.25/min (cost ~\$0.036 → 4–7× markup). + Add-ons: extra number \$5/mo (cost \$1), video avatar \$99/mo, voice biometrics \$49/mo, extra agent \$19/mo.			

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Profit projection — 100 → 5,000 clients

Assumptions: blended ARPU (subscription + minute markup) ≈ \$220/client/mo, Stripe fee ~3%. Ye illustrative model hai — actual usage/churn pe depend karta hai.

CLIENTS	REVENUE/MO	INFRA+USAGE COST	STRIPE 3%	NET PROFIT/MO	NET/YEAR	MARGIN
100	\$22,000	\$5,700	\$660	~\$15,640	~\$188K	71%
500	\$110,000	\$27,900	\$3,300	~\$78,800	~\$946K	72%
1,000	\$220,000	\$56,240	\$6,600	~\$157,160	~\$1.89M	71%
2,500	\$550,000	\$140,000	\$16,500	~\$393,500	~\$4.7M	72%
5,000	\$1,100,000	\$277,750	\$33,000	~\$789,250	~\$9.5M	72%

💰 **BYOK-heavy variant:** agar zyādaतर clients BYOK karein (owner AI cost na bhare), owner cost girkar 100 clients pe ~\$2.5K, 1000 pe ~\$22K ho jaata hai → net margin 85–90% tak (net/year 1000 clients pe ~\$2.3M+). Mix rakho: convenience clients managed (revenue↑), cost-conscious BYOK (margin↑).

⚠ In numbers mein salaries, sales & marketing, taxes shamil nahi — ye pure infra+platform economics hai. Churn, actual minutes, aur tier-mix se real numbers vary karenge. Conservative rakha gaya hai.

8

Smart & innovative monetization — max profit + renewals

Ye meri (analyst) recommendations — margin badhane, churn ghatane aur renewal pakka karne ke liye.

- 1

BYOK-first pricing: Default plan mein client apni AI keys laaye → aap sirf platform fee lo (90% margin), AI cost client pe. Managed minutes ko "convenience" premium banao. Sabse bada margin lever.
- 2

Minute markup 4–7x: Managed minutes cost \$0.036 → charge \$0.15–0.25/min. Included bucket khatam → overage pe profit. Usage-based revenue apne aap scale karta hai.
- 3

Annual billing (2 months free): Upfront cash + **lock-in** → churn kam, cash-flow strong. Monthly se annual pe push karo discount se.
- 4

Reseller revenue-share: Agencies ko white-label do (\$1,999/mo) + unke sub-tenant revenue ka 10–20% lo. Aap bina sales kiye scale karte ho — resellers aapke liye bech'te hain.
- 5

Add-on marketplace: Video avatar, voice biometrics, premium voices, extra numbers — sab paid add-ons (feature already built). High-margin upsell, base plan sasta rakh ke.
- 6

Outcome-based tier: "Pay per booked appointment / qualified lead" — client ke ROI se aligned, premium pricing justify. (Revenue-attribution feature already hai.)
- 7

Setup/onboarding fee: Ek-baar \$299–999 "done-for-you agent setup" — instant cash + client committed (invested → renew karta hai).
- 8

Freemium funnel: Free trial (100 min) → usage dashboard ROI dikhaye → paid conversion. Low CAC.
- 9

Renewal lock-in via value: Call recordings + analytics + trained agents + workflows = high switching cost. ROI dashboard (calls→revenue) har mahine dikhao → renewal obvious.
- 10

Cost optimization: Self-host LiveKit + voice on your servers (scale pe Cloud fee bachao), Telnyx/Plivo over Twilio, Gemini Flash for bulk, R2 zero-egress recordings — margin +10–15%.

✓

Final recommendation

Best combination for max profit: **Balanced stack (Combo 3, \$0.036/min)** as default + **BYOK-first pricing** (high margin) + **managed minutes as premium upsell (4–7x markup)** + **reseller/white-label layer** (scale without sales) + **annual billing** (retention). Isse 1,000 clients pe **~\$1.9M/year net** (managed-mix) se **~\$2.3M+/year** (BYOK-heavy) tak — 71–90% margin. Telephony Telnyx/Plivo, LLM Gemini/Haiku for volume, ElevenLabs/Cartesia voice, LiveKit self-host at scale.